

FULL NAME:

NETID:

CSC 380 Midterm Exam

- *Exam Duration: 75 minutes.*
- *There are 6 questions.*
- *Provide your answers **only inside the boxes** underneath the questions.*
- *Provide neat and concise solutions.*
- *You are allowed to use one sheet (letter or similar size) of your notes.*
- *You are allowed to use a simple four-function calculator.*
- *Good luck!*

Question 1 [20pts]: Mark TRUE or FALSE and briefly explain with one or two sentences. Note that you will receive **half of the points** if your answer is correct, but your explanation is not or if your explanation is missing.

- (a) An unbiased estimator is always better than a biased one.

TRUE FALSE

False. An unbiased estimator is not always better than a biased one; it depends on other factors like variance. Sometimes, a biased estimator with lower variance may be preferable if it results in lower overall error.

- (b) If two events A, B are disjoint then they must be independent.

TRUE FALSE

False. Disjoint (mutually exclusive) events cannot happen together, but that does not imply independence. In fact, if events are disjoint and have positive probabilities, they are dependent because the occurrence of one affects the probability of the other.

- (c) Let X represent the value on a fair six-sided die toss. We have $E(X) = 3.5$.

TRUE FALSE

True. For a fair six-sided die, the expected value $E(X)$ is calculated as the average of all possible outcomes:

$$E(X) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5.$$

- (d) The correlation between a pair of random variables is always between 0 and 1.

TRUE FALSE

False. Correlation between two random variables can range from -1 to 1. A correlation of 0 indicates no linear relationship, while -1 and 1 indicate perfect negative and positive linear

relationships, respectively.

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- (e) Chebyshev's inequality states that for a large sample, the sample mean is approximately normally distributed.

TRUE FALSE

False. Chebyshev's inequality states that for any sample, regardless of the distribution, a certain proportion of observations will lie within a specified number of standard deviations from the mean. It does not imply that the sample mean is approximately normally distributed. The Central Limit Theorem, not Chebyshev's inequality, is what implies that the sample mean approaches a normal distribution for large samples.

Question 2 [15pts]:

Let $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ and let $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i$. Find the mean squared error of this estimator.

Answer:

Solution:

To find the mean squared error (MSE) of the estimator $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i$ for a Poisson distribution, we proceed as follows:

1. Expected Value of the Estimator:

Since $X_i \sim \text{Poisson}(\lambda)$, we know that $\mathbb{E}[X_i] = \lambda$. The estimator $\hat{\lambda}$ is the sample mean:

$$\mathbb{E}[\hat{\lambda}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \cdot n \cdot \lambda = \lambda.$$

So, $\hat{\lambda}$ is an unbiased estimator of λ .

2. Variance of the Estimator:

The variance of the sample mean for independent X_i 's is:

$$\text{Var}(\hat{\lambda}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n \cdot \lambda = \frac{\lambda}{n}.$$

3. Mean Squared Error (MSE):

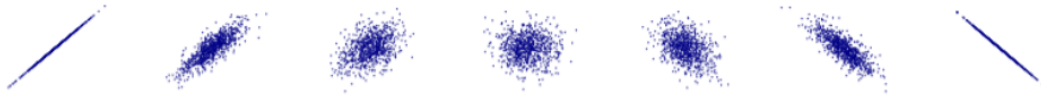
For an unbiased estimator, the mean squared error is simply the variance. Therefore,

$$\text{MSE}(\hat{\lambda}) = \text{Var}(\hat{\lambda}) = \frac{\lambda}{n}.$$

Answer: The mean squared error of the estimator $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i$ is $\frac{\lambda}{n}$.

Question 3 [15pts]:

- (a) Associate each plot below with one of the Pearson correlation coefficient values: -0.4, 0.8, -0.8, -1, 1, 0, 0.4.



Answer: 1, 0.8, 0.4, 0, -0.4, -0.8, -1

- (b) Describe what bias-variance tradeoff means and the relationship between this tradeoff and the mean squared error.

Answer:

Crucially, bias and variance cannot be minimized simultaneously: reducing bias increases variance and vice versa.

$$\text{MSE} = \text{Bias}^2 + \text{Variance}$$

- (c) Describe why we need multiple testing correction and a method to resolve it.

Answer:

Multiple testing correction is essential because performing many statistical tests increases the chance of obtaining false positives. One common method to address this is the Bonferroni correction.

Question 4 [20pts]:

- (a) Suppose that a baseball hitter has a probability of success $p = 0.7$. What is the probability that she hits (has success) more than 6 times out of a total of 15 throws. You may make use of the following outputs from `scipy.stats`: $\text{binom.pmf}(5, 15, 0.7) = 0.003$, $\text{binom.cdf}(5, 15, 0.7) = 0.004$, $\text{binom.pmf}(6, 15, 0.7) = 0.012$, $\text{binom.cdf}(6, 15, 0.7) = 0.015$.

Answer:

To find $P(X > 6)$, we use the complementary probability:

$$P(X > 6) = 1 - P(X \leq 6)$$

Given $P(X \leq 6) = 0.015$:

$$P(X > 6) = 1 - 0.015 = 0.985$$

So, the probability is **0.985**.

- (b) Suppose that a balanced die is rolled. Let A be the event that an even number is obtained, and let B be the event that one of the numbers 1, 2, 3, or 4 is obtained. Prove or disprove whether the events A and B are independent.

Answer:

To check if events A (rolling an even number) and B (rolling 1, 2, 3, or 4) are independent, we need $P(A \cap B) = P(A) \cdot P(B)$.

- Calculate Probabilities: $P(A) = \frac{3}{6} = \frac{1}{2}$, $P(B) = \frac{4}{6} = \frac{2}{3}$, $P(A \cap B) = \frac{2}{6} = \frac{1}{3}$

- Check Independence:

$$P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$$

Since $P(A \cap B) = P(A) \cdot P(B)$, A and B are independent.

Question 5 [15pts]: We take a random sample of size 100 from a distribution with unknown mean μ and with standard deviation $\sigma = 5$. Assume the sample mean is 9. What is the 90% confidence interval for mean μ ? You may make use of the following outputs from `scipy.stats`: $\text{norm.ppf}(0.995) = 2.58, \text{norm.ppf}(0.95) = 1.64, \text{norm.cdf}(0.995) = 0.84, \text{norm.cdf}(0.95) = 0.83$.

Answer:

Given:

- Sample size, $n = 100$
- Sample mean, $\bar{x} = 9$
- Population standard deviation, $\sigma = 5$
- Confidence level = 90%

1. **Calculate the standard error of the mean (SEM):**

$$\text{SEM} = \frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{100}} = \frac{5}{10} = 0.5$$

2. **Determine the z-score for a 90% confidence level:**

For a 90% confidence interval, we need the z-score that corresponds to the middle 90% of the distribution. From the information provided, we have:

$$\text{norm.ppf}(0.95) = 1.64$$

This z-score represents the critical value for a 90% confidence level.

3. **Calculate the confidence interval:**

The formula for the confidence interval is:

$$\bar{x} \pm z \times \text{SEM}$$

Plugging in the values:

$$9 \pm 1.64 \times 0.5$$

$$9 \pm 0.82$$

So, the 90% confidence interval for the mean μ is:

$$(9 - 0.82, 9 + 0.82) = (8.18, 9.82)$$

Answer: The 90% confidence interval for the mean μ is (8.18, 9.82).

Question 6 [15pts]: Suppose that the number of medical in-patient days in nursing homes has a normal distribution with unknown mean μ and unknown variance. Assume we took a random sample of 16 such numbers. The mean of the sample is 180 and the sample standard deviation is 40. We wish to test the following hypotheses: $H_0 : \mu \geq 200$ and $H_A : \mu < 200$. Find out whether we need to reject H_0 at significance 0.01? You may make use of the following outputs from scipy.stats: $t.ppf(0.01, 15) = -2.60, t.cdf(0.01, 15) = 0.50, t.ppf(0.99, 15) = 2.60, t.cdf(0.99, 15) = 0.83$

Answer:

1. **Calculate the test statistic:**

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{180 - 200}{40/\sqrt{16}} = \frac{-20}{10} = -2$$

2. **Critical value:** For a left-tailed test with $\alpha = 0.01$ and $df = 15$, the critical value is $t_{0.01,15} = -2.60$.

3. **Decision:** Since $t = -2$ is not less than -2.60 , we do not reject H_0 at the 0.01 significance level.

Conclusion: There is insufficient evidence to conclude that the mean μ is less than 200.